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Light emitting module

Abstract

A light emitting module includes: a substrate; a plurality of light sources fixed on the substrate and generating individually controllable outputs; a light shielding member disposed between the light sources; a first lens that is disposed above the light sources and on which the light emitted from each of the light sources becomes incident; and a drive unit capable of rotating the substrate. The light sources include a first light source and a second light source. The angle formed by the axis of rotation of the substrate and the optical axis of first light emitted by the first light source and exiting the first lens differs from the angle formed by the axis of rotation and the optical axis of second light emitted by the second light source and exiting the first lens. The second light can irradiate a first circular track having the axis of rotation as its center.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is a national stage of PCT Application No. PCT/JP2022/044189, filed on Nov. 30, 2022, which claims priority to Japanese Application No. 2022-011251, filed on Jan. 27, 2022.

BACKGROUND

Technical Field

(2) The present disclosure relates to a light emitting module.

Background Art

(3) Patent Document 1 below discloses a lighting device that includes a plurality of semiconductor light emitting elements, a casing that holds the semiconductor light emitting elements such that the optical axes of the light emitted from the semiconductor light emitting elements are oriented in the same direction, and a casing driving means for changing the position of the casing along a plane that intersects the optical axes. Rotating the casing around the axial center extending in the direction orthogonal to the plane described above at the center of the semiconductor light emitting elements can mix the light emitted from the semiconductor light emitting elements, thereby eliminating nonuniformity in the color temperature and lighting attributable to individual differences of the semiconductor light emitting elements. The light distribution pattern of this lighting device is constant.

CITATION LIST

Patent Literature

(4) Patent Document 1: JP 2005-121872 A

SUMMARY

Technical Problem

(5) An object of the present disclosure is to provide a light emitting module capable of changing the light distribution pattern.

Solution to Problem

(6) A light emitting module according to one embodiment of the present disclosure includes: a substrate; a plurality of light sources fixed on the substrate and generating individually controllable outputs; a light shielding member disposed between the light sources; a first lens disposed above the light sources and on which the light emitted by each of the light sources becomes incident; and a drive unit capable of rotating the substrate. The light sources include a first light source and a second light source. The angle formed by the axis of rotation of the substrate and the optical axis of first light emitted by the first light source and exiting the first lens differs from the angle formed by the axis of rotation and the optical axis of second light emitted by the second light source and exiting the first lens, allowing the second light to irradiate a first circular track having the axis of rotation as its center.

Advantageous Effects of Invention

(7) According to an embodiment of the present disclosure, a light emitting module capable of changing the light distribution pattern can be provided.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a top view of a light emitting module according to a first embodiment.

(2) FIG. **2** is a partial cross-sectional view taken along line II-II in FIG. **1**.
(3) FIG. **3** is a cross-sectional view enlarging a portion of the cross section shown in FIG. **2**.
(4) FIG. **4** is a cross-sectional view showing a portion of the cross section taken along line IV-IV in FIG. **1**.
(5) FIG. **5** is a cross-sectional view of a portion of the cross section taken along line V-V in FIG. **1**.
(6) FIG. **6**A is a top view of the substrate, the light sources, and the light shielding member shown in FIG. **1**.
(7) FIG. **6**B is a bottom view enlarging one of the light sources in FIG. **6**A.
(8) FIG. **7** is a schematic diagram of an irradiated plane showing the irradiating regions of the light emitted by the light sources and exiting the lens.
(9) FIG. **8**A is a cross-sectional view showing a portion of a light emitting module according to a second embodiment.
(10) FIG. **8**B is a top view of the substrate, the light sources, and the light shielding member shown in FIG. **8**A.
(11) FIG. **9** is a cross-sectional view showing a portion of a light emitting module according to a third embodiment.
(12) FIG. **10** is a cross-sectional view showing a portion of a light emitting module according to a fourth embodiment.
(13) FIG. **11** is a cross-sectional view of the light emitting module shown in FIG. **10** when a cover member is provided on the module.
(14) FIG. **12** is a cross-sectional view showing a portion of a light emitting module according to a fifth embodiment.
(15) FIG. **13** is a cross-sectional view showing a portion of a light emitting module according to a sixth embodiment.
(16) FIG. **14**A is a top view of a variation of the light shielding member.
(17) FIG. **14**B is a top view of a variation of the layout of the light sources.
(18) FIG. **15**A is a top view of a variation of the layout of the light sources.
(19) FIG. **15**B is a top view of a variation of the layout of the light sources.
(20) FIG. **16**A is a top view of a variation of the shape and layout of the light sources.
(21) FIG. **16**B is a bottom view enlarging one of the light sources in FIG. **16**A.
(22) FIG. **17**A is a top view of a variation of the layout of the light sources.
(23) FIG. **17**B is a schematic diagram of an irradiated plane showing the irradiating regions of the light emitted by the light sources and exiting the lens.
(24) FIG. **18**A is a top view of a variation of the wavelength conversion member.
(25) FIG. **18**B is a top view of a variation of the wavelength conversion member.
(26) FIG. **19**A is a top view of a variation of the wavelength conversion member.
(27) FIG. **19**B is a top view of a variation of the wavelength conversion member.
(28) FIG. **20** is a partial cross-sectional view of a variation of the form of securing the lens.
(29) FIG. **21** is a schematic diagram of a variation of the method of controlling the light sources.

DETAILED DESCRIPTION

(30) Certain embodiments of the present disclosure will be explained below with reference to the accompanying drawings.
(31) The drawings are schematic or conceptual. As such, the relationship between the thickness and the width of a portion or part, the dimensional ratio of portions, and the like are not necessarily the same as those in an actual product. Depending on the drawings, the same portion might be depicted with different dimensions or ratio. An end face view only showing a cut section might be used as a cross-sectional view.
(32) In the present specification and the drawings, the same reference numerals denote similar elements that have already been described for which detailed explanation will be omitted as appropriate.

(33) In the present specification, an XYZ orthogonal coordinate system is employed to describe the layout and structure of each part or portion to make the description easily understood. The X, Y, and Z axes are orthogonal to one another. The direction in which the X-axis extends is referred to as “X direction,” the direction in which the Y-axis extends is referred to as “Y direction,” and the direction in which the Z-axis extends will be referred to as “Z direction.” The X direction is an example of a first direction, and the Y direction is an example of a second direction. The X direction in the direction pointed by the arrow will also be referred to as the “+X direction” or the “+X side,” and the X direction going against the arrow will be referred to as the “−X direction” or the “−X side.” Similarly, the Y direction in the direction pointed by the arrow will also be referred to as the “+Y direction” or the “+Y side,” the Y direction going against the arrow will be referred to as the “−Y direction” or the “−Y side,” the Z direction in the direction pointed by the arrow will be referred to as the “+Z direction,” the “+Z side,” or the “upward” direction, and the Z direction going against the arrow will be referred to as the “−Z direction,” the “−Z side,” or the “downward” direction. These directions are irrespective of the direction of gravity. In the embodiments described below, it is assumed that the light sources of the light emitting modules emit light in the +Z direction as an example. Furthermore, the surface of an object viewed from the +Z side is referred to as the “upper face,” and the surface of the object viewed from the −Z side is referred to as the “lower face.”
(34) In the present specification or the scope of claims, moreover, when there are multiple pieces of elements having the same designation and a distinction must be made, a word such as “first,” “second,” or the like might occasionally be added to the element designation. There can be an occasion where a subject distinguished by such a word might differ between that in the present specification and that in the scope of claims. For this reason, even if a constituent designation with the same distinguishing word appear in both the scope of claims and the specification, the subject identified by the word in the scope of claims might not match the subject identified by the word in the specification.
(35) For example, when there are constituent elements distinguished by the words “first,” “second,” and “third” in the present specification, in disclosing the “first” and “third” elements described in the specification in the scope of claims, the elements might be distinguished by using the words, “first” and “second” in the scope of claims from the readability standpoint. In this case, the “first” and “second” elements disclosed in the scope of claims would refer to the “first” and “third” elements described in the present specification. This rule will be reasonably and flexibly applied to not only constituent elements, but also other subjects.
First Embodiment
(36) A first embodiment of the present invention will be explained.
(37) FIG. **1** is a top view of a light emitting module according to this embodiment. FIG. **2** is a partial cross-sectional view taken along line II-II in FIG. **1**. FIG. **3** is a cross-sectional view enlarging a portion of the cross section shown in FIG. **2**. FIG. **4** is a cross-sectional view taken along line IV-IV in FIG. **1**. FIG. **5** is a cross-sectional view taken along line V-V in FIG. **1**. FIG. **6**A is a top view of the substrate, the light sources, and the light shielding member shown in FIG. **1**. FIG. **6**B is a bottom view enlarging one of the light sources in FIG. **6**A. FIG. **7** is a schematic diagram of an irradiated plane showing the irradiating regions of the light emitted by the light sources and exiting the first lens.
(38) In FIG. **2** to FIG. **5**, a portion of the light emitted by each light source **120** is indicated by an arrow. The same applies to the other drawings described later.
(39) A light emitting module **100**, as shown in FIG. **2**, includes a substrate **110**, a plurality of light sources **120**, a first lens **131**, and a drive unit **140**. An example of the use of the light emitting module **100** as a light source of a flashlight of a camera installed in a casing **191** of a smartphone will be described below. The use of the light emitting module **100**, however, is not limited to that described above. Each part of the light emitting module **100** will be described in detail below.

(40) The substrate **110** in this embodiment, as shown in FIG. **3**, is a wiring board having a resin layer **111** and wires **112**. The wires **112** are provided in the substrate **110**.
(41) The surfaces of the substrate **110** include an upper face **110***a* and a lower face **110***b* positioned opposite the upper face **110***a*. The upper face **110***a* and the lower face **110***b* are substantially flat and substantially parallel to the X-Y plane. As shown in FIG. **1**, the top view shape of the substrate **110** is substantially circular. However, the shape of the substrate **110** is not limited to that described above.
(42) As shown in FIG. **3**, the light sources **120** are fixed on the substrate **110**, i.e., on the upper face **110***a* of the substrate **110**. Each light source **120** has a light emitting element **120***a* and a wavelength conversion member **120***b*.
(43) A light emitting element **120***a* is, for example, an LED (light emitting diode). The light emitting element **120***a* includes a semiconductor stack structure **120***c* and a pair of positive and negative electrodes **120***d* and **120***e* disposed on the lower face of the semiconductor stack structure **120***c*. The light emitting element **120***a* may further include a light transmissive substrate or the like disposed on the semiconductor stack structure **120***c*.
(44) The semiconductor stack structure **120***c* includes an n-type semiconductor layer, an active layer, and a p-type semiconductor layer. As shown in FIG. **6**B, the bottom view shape of the semiconductor stack structure **120***c* is quadrangular, in which two opposing sides of the four sides are substantially parallel to the X direction and the remaining two opposing sides are substantially parallel to the Y direction. A light emitting element **120***a* (i.e., a light source **120**) has, for example, a quadrangular shape in a top view, each side being 50 μm to 1000 μm in length. The shape of the semiconductor stack structure **120***c*, however, is not limited to this.
(45) For the materials for the semiconductor stack structure **120***c*, a nitride semiconductor capable of emitting short-wavelength light is preferably used. This allows for efficient excitation of the wavelength conversion substance contained in the wavelength conversion member **120***b*. A nitride semiconductor is primarily expressed by the general formula In.sub.xAl.sub.yGa.sub.1-x-yN (0≤x, 0≤y, x+y≤1). The peak wavelength of the light emitted by the semiconductor stack structure **120***c* is preferably in a range of 400 nm to 530 nm, more preferably 420 nm to 490 nm, even more preferably 450 nm to 475 nm from the standpoint of emission efficiency, excitation of a wavelength conversion substance, and color mixing with the light emitted by the wavelength conversion substance. For the materials for the semiconductor stack structure **120***c*, however, InAlGaAs based semiconductors, InAlGaP based semiconductors, or the like may be used. The color of the light emitted from the semiconductor stack structure **120***c* in this embodiment is blue.
(46) One of the pair of electrodes **120***d* and **120***e* is electrically connected to the n-type semiconductor layer of the semiconductor stack structure **120***c* and the other is electrically connected to the p-type semiconductor layer of the semiconductor stack structure **120***c*. The electrodes **120***d* and **120***e* of the light sources **120** are electrically connected to the wires **112** of the substrate **110** in pairs. Accordingly, the outputs of the light sources **120** are individually controllable.
(47) As the size of a light source **120** decreases, securing the bottom view areas of the electrodes **120***d* and **120***e* becomes more difficult while keeping the pair of electrodes **120***d* and **120***e* apart from one another. In contrast, in this embodiment, as shown in FIG. **6**B, the pair of electrodes **120***d* and **120***e* have right triangle shapes in the bottom view and are arranged in a diagonal direction of the semiconductor stack structure **120***c* such that the right angles of the right triangles oppose two corners of the semiconductor stack structure **120***c*. This can increase the area of each of the pair of electrodes **120***d* and **120***e* in the bottom view while positioning the electrodes adequately apart from one another as compared to the case of a pair of substantially rectangular electrodes whose sides are extending in the X or Y direction, for example. However, the shape of each of the pair of electrodes **120***d* and **120***e*, and the direction of their arrangement are not limited to those described above. For example, in order to easily differentiate the electrode electrically connected to the p-

type semiconductor layer from the electrode electrically connected to the n-type semiconductor layer, the pair of electrodes may have different shapes. Moreover, the pair of electrodes may be arranged along the X or Y direction, and the shape of each electrode may be circular, elliptical, or polygonal such as a quadrangle.
(48) As shown in FIG. **3**, a wavelength conversion member **120***b* is disposed on a light emitting element **120***a*. The wavelength conversion member **120***b* includes a light transmissive resin material as the base material and a wavelength conversion substance. The wavelength conversion member **120***b* may be a sintered body of a wavelength conversion substance.
(49) For the base material of the wavelength conversion members **120***b*, for example, a resin material having light transmissivity such as silicone can be used. The wavelength conversion substance absorbs at least a portion of the primary light emitted by the light emitting elements **120***a* and emits secondary light having a different wavelength from that of the primary light. For the wavelength conversion substance, for example, yttrium aluminum garnet based phosphors (e.g., Y.sub.3(Al,Ga).sub.5O.sub.12:Ce), lutetium aluminum garnet based phosphors (e.g., Lu.sub.3(Al,Ga).sub.5O.sub.12:Ce), terbium aluminum garnet based phosphors (e.g., Tb.sub.3(Al,Ga).sub.5O.sub.12:Ce), CCA-based phosphors (e.g., Ca.sub.10(PO.sub.4).sub.6Cl.sub.2:Eu), SAE based phosphors (e.g., Sr.sub.4Al.sub.14O.sub.25:Eu), chlorosilicate based phosphors (e.g., Ca.sub.8MgSi.sub.4O.sub.16Cl.sub.2:Eu), oxynitride based phosphors, such as β-SiAlON phosphors (e.g., (Si,Al).sub.3(O,N).sub.4:Eu) or α-SiAION phosphors (e.g., Ca(Si,Al).sub.12(O,N).sub.16:Eu), SLA based phosphors (e.g., SrLiAl.sub.3N.sub.4:Eu), nitride based phosphors, such as CASN-based phosphors (e.g., CaAlSiN.sub.3:Eu) or SCASN-based phosphors (e.g., (Sr,Ca)AlSiN.sub.3:Eu), fluoride based phosphors, such as KSF-based phosphors (e.g., K.sub.2SiF.sub.6:Mn), KSAF-based phosphors (e.g., K.sub.2Si.sub.0.99Al.sub.0.01F.sub.5.99:Mn), or MGF-based phosphors (e.g., 3.5MgO.Math.0.5MgF.sub.2.Math.GeO.sub.2:Mn), phosphors having a Perovskite structure (e.g., (CsPb(F,Cl,Br,I).sub.3), quantum dot phosphors (e.g., CdSe, InP, AgInS.sub.2 or AgInSe.sub.2), or the like can be used.
(50) KSAF-based phosphors may have a composition represented by the formula (I) below:
M.sub.2[Si.sub.pAl.sub.qMn.sub.rF.sub.s]   (I)
(51) In the formula (I), M represents an alkali metal, and may include at least K. Mn can be tetravalent Mn ions. P, q, r, and s can satisfy 0.9≤p+q+r≤1.1, 0<q≤0.1, 0<r≤0.2, and 5.9≤s≤6.1, preferably, 0.95≤p+q+r≤1.05 or 0.97≤p+q+r≤1.03, 0<q≤0.03, 0.002≤q≤0.02 or 0.003≤q≤0.015, 0.005≤r≤0.15, 0.01≤r≤0.12 or 0.015≤r≤0.1, 5.92≤s≤6.05 or 5.95≤s≤6.025. Examples of such a composition include the compositions represented by K.sub.2[Si.sub.0.946Al.sub.0.005Mn.sub.0.049F.sub.5.995], K.sub.2[Si.sub.0.942Al.sub.0.008Mn.sub.0.050F.sub.5.992] and K.sub.2[Si.sub.0.939Al.sub.0.014Mn.sub.0.047F.sub.5.986]. Such KSAF-based phosphors can emit high luminance red light having a peak emission wavelength with a narrow full width at half maximum.
(52) The color of the light emitted by the wavelength conversion member **120***b* is yellow, for example. Each light source **120** emits white light as a result of combining the blue light emitted by the light emitting element **120***a* and the yellow light emitted by the wavelength conversion member **120***b*. The color of light emitted by each light source, however, is not limited to white.
(53) The light sources **120** are provided at the intersections of a plurality of first straight lines L**1** extending in the X direction (i.e., the first direction) and arranged in the Y direction (i.e., the second direction) that is orthogonal to the X direction and a plurality of second straight lines L**2** extending in the Y direction and arranged in the X direction. Specifically, as shown in FIG. **6**A, the first straight lines L**1** are arranged in the Y direction at equal intervals. The second straight lines L**2** are arranged in the X direction at equal intervals. In this embodiment, the light sources **120** are

arranged at the 25 intersections of five first straight lines L**1** and five second straight lines L**2**. In other words, 25 light sources **120** are provided in the light emitting module **100**. The center C**2** of each light source **120** in the top view is substantially positioned at an intersection. The center C**2** of the light source **120** located at the center among the 25 light sources **120** is positioned at the center C**1** of the substrate **110** in the top view. As described above, the light sources **120** are arranged in a matrix. The distance between the centers C**2** of the adjacent light sources **120** in the X or Y direction in the top view is, for example, 50 μm to 1000 μm. The distance between the adjacent light sources **120** in the X or Y direction is, for example, 10 μm to 500 μm. The number and positions of the light sources in the light emitting module are not limited to those described above. For example, a light source **120** does not have to be disposed at the center C**1** of the substrate **110**.
(54) As shown in FIG. **3**, the light emitting module **100** according to this embodiment may further include a light shielding member **161** and a light transmissive member **162** that cover the light sources **120**.
(55) The light shielding member **161** in this embodiment, as shown in FIG. **6**A, surrounds the light sources **120** and is disposed between the light sources **120**. The light sources **120** are integrated by the light shielding member **161**. The light shielding member **161** includes, for example, a resin material and a light diffusing material, where the light diffusing material diffuses and reflects the light emitted by the light emitting elements **120***a* and the wavelength conversion member **120***b*. This can reduce the light that exits the lateral faces of the light emitting elements **120***a* without propagating through the wavelength conversion member **120***b*. As a result, the color nonuniformity of the light emitted from the light sources **120** can be reduced. For the resin material included in the light shielding member **161**, a silicone, epoxy, phenol, polycarbonate, or acrylic resin, or their modified resins, or the like can be used. For the light diffusing material contained in the light shielding member **161**, titanium oxide, magnesium oxide, or the like can be used.
(56) The light transmissive member **162** is disposed over and across all of the light sources **120** as shown in FIG. **3**, for example. The light transmissive member **162** can diffuse the light emitted by the light sources **120**. This can make less conspicuous the spaces between the light emitted from adjacent light sources **120** to appear as dark spots. The thickness of the light transmissive member **162** is substantially constant. The light transmissive member **162** is in contact with the upper faces of the light sources **120** and the upper face of the light shielding member **161**. The light transmissive member may be apart from the light sources and the light shielding member. Multiple light transmissive members may be disposed to individually correspond to the light sources **120**. In this case, the light shielding member may be disposed between two adjacent light transmissive members. For the light transmissive member **162**, a resin material, glass, or the like having light transmissivity can be used. The light transmissive member **162** may contain a light diffusing material. For the light diffusing material in the light transmissive member **162**, for example, titanium oxide, magnesium oxide, or the like can be used.
(57) The light emitted from each light source **120** becomes incident on the first lens **131**. The first lens **131** is disposed above and apart from the light transmissive member **162**. The shortest distance between the first lens **131** and the light transmissive member **162** is, for example, 50 μm to 1000 μm. The first lens **131** is a solid of revolution formed around the axis D**1** that passes the center C**1** of the substrate **110** in a top view and extends in the Z direction. Here, a "solid of revolution" means a three-dimensional object formed by rotating a plane around a straight line as an axis while accepting a tolerance in the manufacturing accuracy of the first lens **131**. Accordingly, the axis D**1** constitutes the optical axis of the first lens **131**.
(58) The first lens **131** in this embodiment is a lens having a convex surface protruding towards the light sources **120**. The surfaces of the first lens **131** include a light incident face **131***a* opposing the light sources **120** and an light emission face **131***b* positioned opposite the light incident face **131***a*. The light incident face **131***a* is a convex face. The light emission face **131***b* is flat and substantially parallel to the X-Y plane.

(59) In FIG. **1** and FIG. **6**A, to make the explanation easily understood, the light sources **120** whose centers C**2** are equally distanced from the axis D**1** are indicated by the same hatch or shading patterns. The 25 light sources **120** will be referred to as “first light sources **121**,” “second light sources **122**,” “third light sources **123**,” “fourth light sources **124**,” “fifth light sources **125**,” and “sixth light sources **126**” grouped in the ascending order in terms of the distances from the axis D**1** to the centers C**2**. There is one first light source **121**. There are four second light sources **122**, four third light sources **123**, four fourth light sources **124**, light fifth light sources **125**, and four sixth light sources **126**.
(60) As shown in FIG. **3** to FIG. **5**, the optical axes D**21**, D**22**, D**23**, D**24**, D**25**, and D**26** of the light emitted by each light source **121**, **122**, **123**, **124**, **125**, and **126** that exits the first lens **131** pass through the focal point F**1** of the first lens **131** in this embodiment. The optical axes D**22**, D**23**, D**24**, D**25**, and D**26** of the light emitted by the light sources **122**, **123**, **124**, **125**, and **126** (excluding the first light source **121**) and exiting the first lens **131** become more distant from the axis D**1** in the +Z direction after passing through the focal point F**1**.
(61) Specifically, as shown in FIG. **3**, the angle θ**1** formed by the axis D**1** and the optical axis D**21** of the light emitted by the first light source **121** and exiting the first lens **131** is substantially 0 degrees.
(62) The angle θ**2** formed by the axis D**1** and the optical axis D**22** of the light emitted by any second light source **122** and exiting the first lens **131** is substantially the same because the first lens **131** is a solid of revolution formed around the axis D**1** as a central axis. The angle θ**2** differs from and is greater than the angle θ**1**.
(63) As shown in FIG. **4**, the angle θ**3** formed by the axis D**1** and the optical axis D**23** of the light emitted by any third light source **123** and exiting the first lens **131** is substantially the same because the first lens **131** is a solid of revolution formed around the axis D**1** as the central axis. The angle θ**3** is greater than the angle θ**2**.
(64) As shown in FIG. **3**, the angle θ**4** formed by the axis D**1** and the optical axis D**24** of the light emitted by any fourth light source **124** and exiting the first lens **131** is substantially the same because the first lens **131** is a solid of revolution formed around the axis D**1** as the central axis. The angle θ**4** is greater than the angle θ**3**.
(65) As shown in FIG. **5**, the angle θ**5** formed by the axis D**1** and the optical axis D**25** of the light emitted by any fifth light source **125** and exiting the first lens **131** is substantially the same because the first lens **131** is a solid of revolution formed around the axis D**1** as the central axis. The angle θ**5** is greater than the angle θ**4**.
(66) As shown in FIG. **4**, the angle θ**6** formed by the axis D**1** and the optical axis D**26** of the light emitted by any sixth light source **126** and exiting the first lens **131** is substantially the same because the first lens **131** is a solid of revolution formed around the axis D**1** as the central axis. The angle θ**6** is greater than the angle θ**5**.
(67) The magnitude of each of the angles θ**2**, θ**3**, θ**4**, θ**5**, and θ**6** can be adjusted by way of adjusting, for example, the curvature of the light incident face **131***a* or the distance of each light source **120** from the axis D**1**.
(68) In the peripheral portion of the first lens **131**, a support portion **132** that extends downwards from the peripheral portion of the first lens **131** is provided. The support portion **132** is integrally formed with the first lens **131**. The support portion **132** in this embodiment has a tubular shape that surrounds the light sources **120** in a top view. However, the shape of the support portion is not limited to a tubular shape. For example, multiple columnar supports may be arranged in the peripheral portion of the lens. Furthermore, the support portion may be composed of a different material from that for the lens. In this case, the support portion does not have to have light transmissivity.
(69) As shown in FIG. **2**, an opening **191***a* is provided in the casing **191** of a smartphone. The first lens **131** in this embodiment is disposed in the opening **191***a*. The support portion **132** is fixed to a

constituent element **192** of the smartphone provided in the casing **191** of the smartphone. Between the support portion **132** and the casing **191**, a sealing member **193** adhering to the support portion **132** and the casing **191** is provided. For the sealing member **191**, for example, an elastic material, such as natural rubber or synthetic rubber, can be used. The shape of the sealing member **193**, as shown in FIG. **1**, is annular. The sealing member **193** can reduce the penetration of dust or liquid through the gap between the first lens **131** and the casing **191**. The position of the sealing member, however, is not limited to that described above as long as it can reduce the penetration of dust or liquid through the gap between the lens and the casing. For example, the sealing member may be disposed in the gap between the main body of the lens and the casing. Furthermore, the lens itself may be fixed to the casing without any support portion.

(70) The drive unit **140** in this embodiment has a motor **141** and a shaft **142** that is connected to the substrate **110** and in coordination with the motor **141**. Driving the motor **141** rotates the shaft **142**. As the shaft **142** rotates, the substrate **110** rotates about the axis of rotation D**3** which parallels the Z axis. The axis of rotation D**3** in this embodiment substantially coincides with the axis D**1**.

(71) The light emitting module **100** further includes a control unit **150** capable of controlling the outputs of the light sources **120** in coordination with the drive unit **140**. The shaft **142** is provided with a rotary connector **170**. The rotary connector **170** has a ring unit **171** and a brush unit **172**. The rotary connector **170** electrically connects the wires **112** of the rotating substrate **110** and the control unit **150**. The rotary connector **170** in this embodiment is a slip ring. The rotary connector, however, may be a rotary connector which uses a liquid metal.

(72) The ring unit **171** has a tubular body **171***a* housing and connected to the shaft **142**, and conducting rings **171***b* disposed on the periphery of the tubular body **171***a*. The ring unit **171** rotates with the shaft **142**. A plurality of rings **171***b* and a plurality of wires **112** built into the substrate **110** are electrically connected in pairs via the interior of the shaft **142** and the interior of the tubular body **171***a*.

(73) The brush unit **172** has a plurality of conducting brushes **172***a* contacting the rings **171***b* in pairs, and a holder **172***b* that holds the brushes **172***a*.

(74) The control unit **150** has, for example, a CPU (central processing unit) and a memory. The control unit **150** is electrically connected to the motor **141** of the drive unit **140** and each brush **172***a* of the rotary connector **170**.

(75) The control unit **150** and the brush unit **172** in this embodiment are fixed to the constituent element **192** of the smartphone. Accordingly, when the motor **141** is operated, an electrical signal can be sent to the ring unit **171** without rotating the control unit **150** and the brush unit **172**.

(76) The operation of the light emitting module **100** according to this embodiment will be described next.

(77) The control unit **150** controls the drive unit **140** to rotate the substrate **110**. The rotational speed of the substrate **110** is not particularly limited, but is in a range of 60 rpm to 24000 rpm, for example. The control unit **150** can be adapted to adjust the rotational speed of the motor **141** of the drive unit **140**.

(78) As shown in FIG. **3**, the angle θ**1** formed by the axis of rotation D**3** and the optical axis D**21** of the light emitted by the first light source **121** and exiting the first lens **131** is substantially 0 degrees. Accordingly, when the first light source **121** is lit during a rotation of the substrate **110**, the majority of the light emitted by the first light source **121** and exiting the first lend **131** rotates around the axis of rotation D**3** in the irradiated plane P**1** that is orthogonal to the axis of rotation D**3** and located in the +Z direction of the first lens **131**. In this embodiment, the irradiated plane P**1** is an imaginary plane. The irradiating region of the light emitted by the first light source **121** and exiting the first lens **131** in the irradiated plane P**1** when the first light source **121** is lit during a rotation of the substrate **110** will be referred to as an “irradiating region **121***a*” below. As shown in FIG. **7**, the irradiating region **121***a* is substantially a circular region having the axis of rotation D**3** as its center.

(79) As shown in FIG. **3**, the angle θ**2** formed by the axis of rotation D**3** and the optical axis D**22** of the light emitted by each second light source **122** and exiting the first lens **131** is different from θ**1** and is greater than zero degrees. When a second light source **122** is lit during a rotation of the substrate **110**, the light emitted by the second light source **122** and exiting the first lens **131** rotates around the axis of rotation D**3** in the irradiated plane P**1** in the state in which the optical axis D**22** is oblique to the axis of rotation D**3**. The irradiating region of the light emitted by the second light source **122** and exiting the first lens **131** in the irradiated plane P**1** when the second light source **122** is lit during a rotation of the substrate **110** will be referred to as an “irradiating region **122***a*” below. As shown in FIG. **7**, the irradiating region **122***a* is an annular region located on the outward of the irradiating region **121***a*. A portion of the irradiating region **122***a* may overlap the irradiating region **121***a*. As the substrate **110** rotates, the optical axis D**22** moves on a first circular track **122***b* in the irradiating region **122***a*. The first circular track **122***b* when viewed in the Z direction (from the −Z side in FIG. **3**) is a track having the axis of rotation D**3** as its center.
(80) As shown in FIG. **3** and FIG. **4**, the angle θ**3** formed by the axis of rotation D**3** and the optical axis D**23** of the light emitted by a third light source **123** and exiting the first lens **131** is different from the angle θ**1** and θ**2**, and is greater than θ**2**. When a third light source **123** is lit during a rotation of the substrate **110**, the light emitted by the third light source **123** and exiting the first lens **131** rotates around the axis of rotation D**3** in the irradiated plane P**1** in the state in which the optical axis D**23** is positioned more distant from the axis of rotation D**3** than the optical axis D**22** is. The irradiating region of the light emitted by the third light source **123** and exiting the first lens **131** in the irradiated plane P**1** when the third light source **123** is lit during a rotation of the substrate **110** will be referred to as an “irradiating region **123***a*” below. As shown in FIG. **7**, the irradiating region **123***a* is an annular region in which a portion thereof overlaps the irradiating region **122***a* and the other portion positioned on the outward of the irradiating region **122***a*. As the substrate **110** rotates, the optical axis D**23** moves on a second circular track **123***b* in the irradiating region **123***a*. The second circular track **123***b* when viewed in the Z direction (from the −Z direction in FIG. **3**) is a track having the axis of rotation D**3** as its center.
(81) Similar to the third light source **123**, when each of the light sources **124**, **125**, and **126** is lit during a rotation of the substrate **110**, as shown in FIG. **3**, FIG. **4**, and FIG. **5**, the light emitted by each light source **124**, **125**, and **126** and exiting the first lens **131** rotates around the axis of rotation D**3** in the irradiated plane P**1** in the state in which each optical axe D**24**, D**25**, and D**26** is oblique to the axis of rotation D**3**. The larger the distance between the center C**2** of a light source **120** and the axis of rotation D**3**, the larger the angle formed by the axis of rotation D**3** and the optical axis of the light emitted by the light source **120** and exiting the first lens **131** results. Thus, the angle θ**3**<the angle θ**4**<the angle θ**5**<the angle θ**6**. The irradiating regions of the light emitted respectively by the light sources **124**, **125**, and **126** and exiting the first lens **131** in the irradiated plane P**1** when the light sources **124**, **125**, and **126** are respectively lit during a rotation of the substrate **110** will be referred to as an “irradiating region **124***a*,” “irradiating region **125***a*,” and “irradiating region **126***a*” below.
(82) As shown in FIG. **7**, the irradiating region **124***a* is an annular region in which a portion overlaps the irradiating region **123***a* and the other portion is located on the outward of the irradiating region **123***a*. As the substrate **110** rotates, the optical axis D**24** moves on a third circular track **124***b* in the irradiating region **124***a*. The irradiating region **125***a* is an annular region in which a portion overlaps the irradiating region **124***a* and the other portion is located on the outward of the irradiating region **124***a*. As the substrate **110** rotates, the optical axis D**25** moves on a fourth circular track **125***b* in the irradiating region **125***a*. The irradiating region **126***a* is an annular region in which a portion overlaps the irradiating region **125***a* and the other portion is located on the outward of the irradiating region **125***a*. As the substrate **110** rotates, the optical axis D**26** moves on a fifth circular track **126***b* in the irradiating region **126***a*. The third circular track **124***b*, the furth circular track **125***b*, and the fifth circular track **126***b* when viewed in the Z direction (from the −Z

side in FIG. **3**) are tracks having the axis of rotation D**3** as their center.
(83) The optical axes D**22**, D**23**, D**24**, D**25**, and D**26** of the light emitted by the light sources **122**, **123**, **124**, **125**, and **126**, excluding the first light source **121**, and exiting the first lens **131** become more distant from the axis D**1** in the +Z direction after passing the focal point F**1**. Accordingly, the light emitting module **100** can irradiate a broader area of the irradiated plane P**1** while being reduced in the module size.
(84) In this embodiment, the radii of the circular tracks **122***b*, **123***b*, **124***b*, **125***b*, and **126***b* respectively correspond to the distances from the respective centers C**2** of the light sources **122**, **123**, **124**, **125**, and **126** to the axis of rotation D**3**. In this embodiment, because the light sources **120** are arranged in a matrix, and the centers C of the light sources **120** are not positioned on equally spaced circumferences around the axis of rotation D**3**, the spacing of adjacent circular tracks is not uniform. However, even when the light sources **120** are arranged in a matrix, the circular tracks **122***b*, **123***b*, **124***b*, **125***b*, and **126***b* can be arranged at substantially equal intervals by adjusting the shape of the first lens **131**.
(85) In this embodiment, moreover, adjacent irradiating regions among the irradiating regions **122***a*, **123***a*, **124***a*, **125***a*, and **126***a* partly overlap each other. Allowing adjacent regions to have overlapping portions can reduce the occurrence of a low brightness region between adjacent circular tracks. However, depending on the application of the light emitting module **100**, the irradiating regions may be set such that no two adjacent regions overlap each other. In this embodiment, furthermore, the irradiating region **121***a* is solid circular, the irradiating regions **122***a*, **123***a*, **124***a*, **125***a*, and **126***a* are annular, and the circular tracks **122***b*, **123***b*, **124***b*, **125***b*, and **126***b* are annular. However, the irradiating regions do not have to be strictly circular or annular depending on the manufacturing accuracy of each part or assembly accuracy of parts. The circular tracks do not have to be strictly circular depending on the manufacturing accuracy of each part or assembly accuracy of parts.
(86) For example, when there are multiple subjects in a photographing region of a camera at different distances from the light emitting module **100**, and light of substantially uniform luminous intensity irradiates the entire photographing region, the brightness of the subject declines as the distance of the subject from the light emitting module **100** increases. For this reason, when the camera captures an image of multiple subjects, a nearby subject might be clipped to appear white and a distant subject might be clipped to appear black in the captured image. In contrast, the control unit **150** in this embodiment individually controls the outputs of the light sources **120** while the substrate **110** rotates.
(87) Specifically, the control unit **150** controls the output of the first light source **121** such that the brightness of the irradiating region **121***a* achieves predetermined brightness. This, for example, can achieve the brightness of the irradiating region **121***a* in accordance with the distance of a subject located in the irradiating region **121***a* viewed in the Z direction. At this time, the control unit **150** may be adapted to set the output of the light source **121** by further incorporating a condition into the drive conditions of the light emitting module **100** such as the rotational speed or the number of rotation of the substrate **110** and/or the shooting conditions such as the shutter speed of the camera.
(88) Moreover, the control unit **150** controls the outputs of the light sources excluding the first light source **121**, i.e., the light sources **122**, **123**, **124**, **125**, and **126**, in coordination with the rotation of the substrate **110**. Specifically, for example, the control unit **150** controls the outputs of the light sources **122**, **123**, **124**, **125**, and **126** in accordance with the positions of the optical axes D**22**, D**23**, D**24**, D**25**, and D**26** on the respective circular tracks **122***b*, **123***b*, **124***b*, **125***b*, and **126***b*. This, for example, can make the circumferential brightness distribution of the irradiating region **122***a* in accordance with the distance in the Z direction to each subject located in the irradiating region **122***a*. This similarly applies to the other irradiating regions **123***a*, **124***a*, **125***a*, and **126***a*. At this time, the control unit **150** may be adapted to set the outputs of the light sources **122**, **123**, **124**, **125**, and **126** by further incorporating a condition into the drive conditions of the light emitting module

100, such as the rotational speed or the number of rotation of the substrate **110**, and/or the shooting conditions such as the shutter speed of the camera. The angle θ**1** formed by the axis of rotation D**3** and the optical axis D**21** of the light emitted by the first light source **121** and exiting the first lens **131** may be larger than 0 degrees. In other words, the optical axis D**21** of the first light source **121** may move on a circular track having the axis of rotation D**3** as its center. In this case, the control unit **150** may be adapted to control the output of the first light source **121** in coordination with the rotation of the substrate **110**. In this manner, light irradiation with a predetermined brightness distribution can be achieved on the irradiating region **121***a*.
(89) In the manner described above, the light distribution pattern of the light emitting module **100** can be changed.
(90) The control unit **150**, for example, may be adapted to estimate the position of the optical axis of the light emitted by each light source **120** and exiting the first lens **131** on the circular track from the position of the light source **120** before rotation and the rotational speed, the number of rotation, or the like of the motor **141**. The control unit **150** may be adapted to estimate the position of the optical axis of the light emitted by each light source **120** and exiting the first lens **131** on the circular track by using a detection result of a rotational angle detection sensor such as a rotary encoder. Specifically, a rotary angle detection sensor detects the rotational angle of the substrate **110** from a reference position in the state in which the substrate **110** is not rotating. The position of the optical axis of the light emitted by each light source **120** and exiting the first lens **131** on the circular track during rotation can be estimated based on the rotational angle of the substrate **110**.
(91) The effect of this embodiment will be described next.
(92) A light emitting module **100** according to this embodiment includes a substrate **110**, a plurality of light sources **120** secured on the substrate **110** and generating individually controllable outputs, a first lens **131** provided above the light sources and on which the light emitted from the light sources **120** becomes incident, and a drive unit **140** capable of rotating the substrate **110**. The light sources **120** include a first light source **121** and a second light source **122**. The angle θ**1** formed by the axis of rotation D**3** of the substrate **110** and the optical axis D**21** of the light emitted by the first light source **121** and exiting the first lens **131** differs from the angle θ**2** formed by the axis of rotation D**3** and the optical axis D**22** of the light emitted by the second light source **122** and exiting the first lens **131**. The second light emitted by the second light source **122** and exiting the first lens **131** can irradiate a first circular track **122***b* having the axis of rotation D**3** as its center. Accordingly, the light distribution pattern of the light emitting module **100** can be changed.
(93) Furthermore, because the light from each light source **120** becomes incident on a single first lens **131**, the light emitting module **100** of this embodiment is structurally simple and compact as compared to a light emitting module having lenses that individually correspond to multiple light sources **120**. The light emitting module has good design quality because of a single first lens **131** that is visually recognizable from the outside. Because multiple light sources **120** are arranged under a single first lens **131**, the light sources **120** can be easily arranged close together. This makes it easy to increase the number of irradiating regions **121***a*, **122***a*, **123***a*, **124***a*, **125***a*, and **126***a* whose brightness can be individually controlled while reducing the size of the light emitting module **100**.
(94) Furthermore, the light sources include a third light source **123**, and the third light emitted by the third light source **123** and exiting the first lens **131** can irradiate a second circular track **123***b* around the axis of rotation D**3** that is different from the first circular track **122***b*. In other words, there are two or more brightness-controllable irradiating regions i.e., **121***a*, **122***a*, and **123***a*. This can increase the light distribution pattern variations for the light emitting module **100**.
(95) Furthermore, the first lens **131** is separated from the substrate **110**. In other words, the first lens **131** does not rotate with the substrate **110**. Because the first lens **131** overlaps the substrate **110** in the Z direction, the first lens **131** can reduce the external exposure of the substrate **110** and light sources **120** that rotate.
(96) Moreover, the first lens **131** is a solid of revolution formed around the axis of rotation D**3**

which coincides with the optical axis of the first lens **131**. This can simplify the shape of the first lens **131**. Providing multiple second light sources **122** under such a first lens **131** at substantially the same distance from the axis of rotation D**3** allows a single irradiating region **122***a* to be irradiated by the multiple second light sources **122**. This can improve the brightness of the irradiating region **122***a*.
(97) Furthermore, the light emission face **131***b* of the first lens **131** in this embodiment is flat. Accordingly, it is more difficult for foreign matter to remain on the light emission face **131***b* than an uneven light emission face. This eliminates the need for a cover member. Not having a cover member can lessen the luminous intensity decline in the light emitted from the light emitting module **100**.
(98) Moreover, the light incident face **131***a* of the first lens **131** is convex. Thus, the light emitted by each light source **120** and exiting the first lens **131** advances towards the optical axis of the first lens **131**. This can reduce the blocking of the light exiting the first lens **131** by the peripheral parts of the light emitting module **100** such as the casing **191**.
(99) The light sources **120** are arranged at the intersections of the first straight lines L**1** extending in the X direction and arranged in the Y direction orthogonal to the X direction and the second straight lines L**2** extending in the Y direction and arranged in the X direction in a top view. A matrix arrangement of the light sources **120** like this facilitates the arrangement of a plurality of light sources **120** on the substrate **110**.
(100) Furthermore, a light shielding member **161** is disposed between the light sources **120**. This can reduce color nonuniformity of the light emitted from the light sources **120**.
(101) The light emitting module **100** further includes a light transmissive member **162** disposed over and across all of the light sources **120**. This can make less conspicuous the spaces between the light emitted from adjacent light sources **120** to appear as dark spots.
(102) Moreover, the light transmissive member **162** contains a light diffusing material. This can suitably make less conspicuous the spaces between the light emitted from adjacent light sources **120** to appear as dark spots.
Second Embodiment
(103) A second embodiment will be described next.
(104) FIG. **8**A is a cross-sectional view of a portion of a light emitting module according to this embodiment. FIG. **8**B is a top view of the substrate, the light sources, and the light shielding member shown in FIG. **8**A.
(105) A light emitting module **200** according to this embodiment differs from the light emitting module **100** according to the first embodiment such that it has nine light sources **120**.
(106) In the description below, the differences from the first embodiment will be focused basically, and similar elements to those in the first embodiment might be omitted. The same applies to the other embodiments and variations discussed later.
(107) As shown in FIG. **8**B, the light sources **120** are arranged at the intersections of three first straight lines L**1** and three second straight lines L**2**. The center C**2** of the light source **121** among the light sources **120** that is located furthest on the +X side and furthest on the −Y side is positioned on the axis of ration D**3**. Such a light emitting module **200**, similar to the first embodiment, can also have six irradiating regions the brightness of which can be individually controlled. The plurality of irradiating regions whose brightness is individually controllable is not limited to this, and can be modified by altering the number or the layout of the light sources.
Third Embodiment
(108) A third embodiment will be described next.
(109) FIG. **9** is a cross-sectional view of a portion of a light emitting module according to this embodiment.
(110) A light emitting module **300** according to this embodiment differs from the light emitting module **100** according to the first embodiment in terms of the shape of the first lens **131**.

(111) The first lens **331** is a Fresnel lens. The light incident face **331***a* of the first lens **331** has a Fresnel profile. Here, a "Fresnel profile" refers to a sawtooth cross section. The light emission face **331***b* of the first lens **331** is flat and substantially parallel to the X-Y plane.
(112) Similar to the first embodiment, the angles formed by the axis of rotation D**3** and the optical axes of the light emitted by the light sources **121**, **122**, **123**, **124**, **125**, and **126** and exiting the first lens **331** in this embodiment differ from one another.
(113) The effect of this embodiment will be explained next.
(114) The first lens **331** is a Fresnel lens in which the light incident face **331***a* has a Fresnel profile. The first lens **331** which is a Fresnel lens, for example, is a lens in which the curved face of the first lens **131** is divided into concentric annular sections, each section being angled and reduced in thickness. Furthermore, the first lens **331** externally visible being a Fresnel lens can improve the design quality.
Fourth Embodiment
(115) A fourth embodiment will be described next.
(116) FIG. **10** is a cross-sectional view of a portion of a light emitting module according to this embodiment. FIG. **11** is a cross-sectional view of the light emitting module shown in FIG. **10** on which a cover member is provided.
(117) A light emitting module **400** according to this embodiment differs from the light emitting module **100** according to the first embodiment in terms of the shape of the first lens **431**.
(118) The first lens **431** is a convex lens. Both the light incident face **431***a* and the light emission face **431***b* of the first lens **431** are convex faces. In this embodiment, the light incident face **431***a* and the light emission face **431***b* of the first lens **431** are asymmetrical. Specifically, the top view shapes of the light incident face **431***a* and the light emission face **431***b* are both circular, and the diameter of the light incident face **431***a* is larger than the diameter of the light emission face **431***b* in the top view. Furthermore, the curvature of the light incident face **431***a* is smaller than the curvature of the light emission face **431***b*. When the curvature of the light incident face of the first lens is small, the light from the light sources becomes incident on the first lens efficiently. When the curvature of the light emission face is large, stray light is reduced and contrast is improved. However, the shapes of the light incident face and the light emission face are not limited to those described above. For example, the curvature of the light incident face of the first lens may be larger than the curvature of the light emission face. In this case, the light from the light sources can be refracted and bent greatly by the first lens to achieve a wide angle light distribution. The light incident face and the light emission face may be substantially symmetrical in the Z direction.
(119) The majority of the light emitted by each light source **120** refracted and become incident on the light incident face **431***a* is refracted again at the light emission face **431***b*. Similar to the first embodiment, the angles formed by the axis of rotation D**3** and the optical axes of the light emitted by the light sources **121**, **122**, **123**, **124**, **125**, and **126** and exiting the first lens **431** differ from one another.
(120) The effect of this embodiment will be explained next.
(121) The light incident face **431***a* and the light emission face **431***b* of the first lens **431** are both convex faces. As such, the angle formed by the axis of rotation D**3** and the optical axis of the light emitted by each light source **120** and exiting the first lens **431** can be easily adjusted by way of adjusting the shapes of the two convex faces, i.e., the light incident face **431***a* and the light emission face **431***b*.
(122) As shown in FIG. **11**, a cover member **494** having light transmissivity may be provided above the first lens **431**. This can reduce the chance of allowing the convex light emission face **431***b* to be externally exposed to be bumped and damaged. At this time, a sealing member **193** does not have to be disposed.
Fifth Embodiment
(123) A fifth embodiment will be described next.

(124) FIG. **12** is a cross-sectional view of a portion of a light emitting module according to this embodiment.
(125) A light emitting module **500** according to this embodiment differs from the light emitting module **400** according to the fourth embodiment by having no transmissive member **162** and further including multiple second lenses **580**.
(126) The second lenses **580** are individually disposed on the light sources **120**. Each second lens **580** is a convex lens, for example. The second lenses **580** substantially have the same shape. However, the shapes of the second lenses do not have to be identical. Each second lens may be apart from the corresponding light source. The shortest distance between the first lens **431** and the second lenses **580** is, for example, 50 μm to 1000 μm.
(127) The light emitted by each light source **120** becomes incident on the corresponding second lens **580**. The light exiting each second lens **580** becomes incident on the first lens **431**.
(128) The effect of this embodiment will be explained next.
(129) The light emitting module **500** according to this embodiment further includes second lenses **580** located between the first lens **431** and the light sources **120**, and disposed on the light sources **120**. This allows the second lenses **580** to collect the light from the light sources **120** to be incident on the first lens **431**.
(130) Furthermore, the angle formed by the axis of rotation D**3** and the optical axis of the light emitted by each light source **120** and exiting the first **431** can be easily finely adjusted by each second lens **580**.
Sixth Embodiment
(131) A sixth embodiment will be described next.
(132) FIG. **13** is a cross-sectional view of a portion of a light emitting module according to this embodiment.
(133) A light emitting module **600** according to this embodiment differs from the light emitting module **500** according to the fifth embodiment such that, in place of multiple second lenses **580**, a single second lens **680** is positioned between the first lens **431** and the light sources **120**, and disposed over and across all of the light sources **120**.
(134) The second lens **680** is, for example, a convex lens. The second lens **680** is in contact with the upper faces of the light sources **120** and the upper face of the light shielding member **161**. However, the second lens may be set apart from the light sources and the light shielding member.
(135) The light emitted by each light source **120** becomes incident on the second lens **680**. The light exiting the second lens **680** becomes incident on the first lens **431**.
(136) The effect of this embodiment will be explained next.
(137) In the light emitting module **600** according to this embodiment, the second lens **680** is disposed across all of the light sources **120**. The light emitting module **600** also allows the second lens **680** to collect the light from each of the light sources **120** before becoming incident on the first lens **431**.
(138) The second lens **680** is disposed over and across all of the light sources **120**. Providing a single second lens **680** like this can reduce the plurality of components of the light emitting module **600**.
(139) Furthermore, the angles formed by the axis of rotation D**3** and the optical axes of the light from the light sources **120** and exiting the first lens **431** may be finely adjusted by using the second lens **680**.
Variations
(140) FIG. **14**A is a top view of a variation of the light shielding member.
(141) As shown in FIG. **14**A, the light shielding members **261** may individually surround the light sources **120**. Light transmissive members may be disposed individually on the wavelength conversion members **120***b* of the light sources **120**, and the lateral faces of the light transmissive members may be covered by the light shielding member **261**.

(142) FIG. **14**B is a top view of a variation of the layout of light sources.
(143) As shown in FIG. **14**B, the light sources **120** do not have to be disposed at the intersections of the first straight lines L**1** and the second straight lines L**2** located at the corners. In other words, no light source **120** is provided at the intersection located furthest on the +X side and furthest on the +Y side, the intersection located furthest on the −X side and furthest on the +Y side, the intersection located furthest on the +X side and furthest on the −Y side, and the intersection located furthest on the −X side and furthest on the −Y side. In such a case, the light sources **120** can be arranged in correspondence with the circle E**1** having the axis of rotation D**3** as its center.
(144) FIG. **15**A is a top view of a variation of the layout of light sources.
(145) As shown in FIG. **15**A, on the substrate **110**, the light sources **120** may be arranged on the circumferences G**1**, G**2**, and G**3** that are arranged around the axis of rotation D**3** and have radii different from one another. In other words, on each of the circumferences G**1**, G**2**, and G**3**, multiple light sources **120** are circumferentially arranged at substantially equal intervals. In such a case, the light sources **120** can be easily grouped by the distances from their centers C**2** to the axis of rotation D**3**, thereby facilitating the design of the first lens **131** and the control applied by the control unit **150**.
(146) The larger the angle formed by the axis of rotation and the optical axis of the light emitted by a light source **120** and exiting the first lens **131**, the faster the circumferential speed of the optical axis of the light source **120** moving on the circular track results. For this reason, the farther the distance of the irradiating region from the axis of rotation, the lower the brightness tends to result. In this variation, the number of light sources **120** that can be arranged increases as the diameter of the circumference increases. By arranging more light sources on a larger diameter circumference, the decline in the brightness of the irradiating region achieved by the light exiting the first lens **131** can be lessened. This can reduce the brightness nonuniformity among the irradiating regions attributable to a circumferential speed difference.
(147) FIG. **15**B is a top view of a variation of the layout of light sources.
(148) As shown in FIG. **15**B, the light sources **120** may be arranged at the intersections of the first straight lines L**1** and the second straight lines L**22**. The first straight line L**1** extend in the X direction and are arranged in the Y direction. The second straight lines L**22** extend in the H**1** direction which intersects the X and Y directions and are arranged in the H**2** direction orthogonal to the H**1** direction. For example, adjacent light sources **120** in the H**1** direction are such that their centers are positioned at one-half of the distance between the centers C**2** of adjacent light sources **120** in the X direction in a top view. In such a case, the light sources **120** can be easily grouped by the distances from their centers C**2** to the axis of rotation D**3**, thereby facilitating the design of the first lens **131** and the control applied by the control unit **150**.
(149) Furthermore, the light sources can be arranged such that the number of light sources whose centers C**2** are more distant from the axis of rotation D**3** is greater. This can lessen the brightness decline in the irradiating region of the light emitted by the light sources **120** whose centers C**2** are more distant from the axis of rotation D**3** and exiting the first lens **131**, thereby reducing the brightness nonuniformity among the irradiating regions attributable to a circumferential speed difference.
(150) FIG. **16**A is a top view of a variation of the light source shape and layout.
(151) FIG. **16**B is a bottom view of a light source shown in FIG. **16**A.
(152) As shown in FIG. **16**A, the shape of each light source **220** may be a hexagon in the top view. The light sources **220** may be arranged in a honeycomb shape. In the case of such an arrangement, the light sources **120** can be easily grouped by the distances from their centers C**2** to the axis of rotation D**3**, thereby facilitating the design of the first lens **131** and the control applied by the control unit **150**.
(153) Furthermore, the light sources **220** can be arranged such that the number of light sources whose centers C**2** are more distant from the axis of rotation D**3** is greater. This can lessen the

brightness decline in the irradiating region of the light emitted by the light source **220** whose centers C**2** are more distant from the axis of rotation D**3** and exiting the first lens **131**, thereby reducing the brightness nonuniformity among the irradiating regions attributable to a circumferential speed difference.

(154) As shown in FIG. **16**B, the pair of electrodes **220***d* and **220***e* of the light emitting element **220***a* are arranged in the Y direction orthogonal to the X direction in which one of the sides of the perimeter extends. The bottom view shape of each of the electrodes **220***d* and **220***e* is a trapezoid. The pair of electrodes **220***d* and **220***e* are arranged such that they have point symmetry relative to the center C**2** of the light source **220**. The shape and the direction of arrangement of the electrodes **220***d* and **220***e* can be suitably changed in accordance with the shape of the light source **220**.

(155) FIG. **17**A is a top view of a variation of the layout of light sources. FIG. **17**B is a schematic diagram of an irradiated plane showing the irradiating regions of the light emitted by the light sources and exiting the lens.

(156) As shown in FIG. **17**A, the light sources **122**, **123**, **124**, **125**, and **126** may be arranged such that their respective centers C**2** are positioned on the circumferences G**21**, G**22**, G**23**, G**24**, and G**25** located at equal intervals around the axis of rotation D**3**. Specifically, the centers C**2** of the second light sources **122** are positioned on the circumference G**21**, the centers C**2** of the third light sources **123** are positioned on the circumference G**22** located outward from the circumference G**21**, the centers C**2** of the fourth light sources **124** are positioned on the circumference G**23** located outward from the circumference G**22**, the centers C**2** of the fifth light sources **125** are positioned on the circumference G**24** located outward from the circumference G**23**, and the centers C**2** of the sixth light sources **126** are positioned on the circumference G**25** located outward from the circumference G**24**. This can substantially space the circular tracks **122***b*, **123***b*, **124***b*, **125***b*, and **126***b* at equal intervals as shown in FIG. **17**B without adjusting the curvature of the first lens **131**. At this time, adjacent irradiating regions among the irradiating regions **122***a*, **123***a*, **124***a*, **125***a*, and **126***a* partly overlap. Adjacent irradiating regions do not have to partly overlap.

(157) FIG. **18**A is a top view of a variation of the wavelength conversion member.

(158) In FIG. **18**A, for ease of understanding, the location where the wavelength conversion member **320***b* is provided is shown with a diagonal hatch pattern.

(159) As shown in FIG. **18**A, a single wavelength conversion member **320***b* may be disposed over and across all of the light emitting elements **120***a*. In such a case, each portion where a light emitting element **120***a* overlaps the wavelength conversion member **320***b* in the Z direction corresponds to a light source **320**.

(160) FIG. **18**B is a top view of a variation of the wavelength conversion member.

(161) In FIG. **18**B, for ease of understanding, the locations where the wavelength conversion members **420***b* are provided are shown with a diagonal hatch pattern.

(162) As shown in FIG. **18**B, multiple annular wavelength conversion members **420***b* may be disposed such that each wavelength conversion member **420***b* is disposed across annularly arranged light emitting elements **120***a*. In this case, each portion where a light emitting element **120***a* overlaps a wavelength conversion member **420***b* in the Z direction corresponds to a light source **420** as well.

(163) FIG. **19**A is a top view of a variation of the wavelength conversion member.

(164) In FIG. **19**A, two types of wavelength conversion members **529**A and **529**B are shown with different diagonal hatch patterns.

(165) As shown in FIG. **19**A, a group of light sources **520**A among the light sources **520** may have wavelength conversion members **529**A emitting light of the chromaticity equivalent to one another and another group of light sources **520**B among the light sources **520** may have wavelength conversion members **529**B emitting light of different chromaticity from that of the light emitted by the wavelength conversion members **529**A. In other words, the chromaticity of the light emitted by the light sources **520**A differs from the chromaticity of the light emitted by the light sources **520**B.

However, the method of varying the chromaticity of multiple light sources is not limited to that described above. For example, the peak wavelength of the light emitted by a light emitting element may be varied.
(166) In FIG. **19**A, the number of the light sources **520**A is the same as that of the light sources **520**B. In a top view, the light sources **520**A and the light sources **520**B are arranged to have line symmetry with respect to the straight line P**2** passing through the axis of rotation D**3** and paralleling the Y-axis. The number of the light sources **520**A does not have to be the same as that of the light sources **520**B.
(167) In such a case, for each light source **520**A, there is a light source **520**B whose center C**2** is equally distanced from the axis of rotation D**3**. Thus, the light emitted by a light source **520**A and exiting the first lens **131** irradiates the same circular track irradiated by the light emitted by the light source **520**B, whose center C**2** is equally distanced from the axis of rotation D**3**, and exiting the first lens **131**. Accordingly, the color of the light emitted by the light source **520**A and exiting the first lens **131** and the color of the light emitted by the light source **520**B and exiting the first lens **131** can be mixed. Furthermore, allowing the control unit **150** to control the output ratio of the light from the light sources **520**A and **520**B can adjust the degree of color mixing. This allows for emission of light that has been tuned to a predetermined color.
(168) FIG. **19**B is a top view of a variation of the wavelength conversion member.
(169) In FIG. **19**B, four types of wavelength conversion members **529**A, **529**B, **529**C, and **529**D are shown with different diagonal hatch patterns.
(170) As shown in FIG. **19**B, four types of wavelength conversion members **529**A, **529**B, **529**C, and **529**D emitting light of different chromaticity from one another may be provided.
(171) In FIG. **19**B, the numbers of light sources **520**A, **520**B, **520**C, and **520**D having wavelength conversion members **529**A, **529**B, **529**C, and **529**D, respectively, are the same. In the top view, moreover, the light sources **520**A are arranged in the region that is located on the +Y side and the −X side among the four regions divided by the straight line P**2** passing through the axis of rotation D**3** and paralleling the Y-axis and the straight line P**3** orthogonal to the straight line P**2**. The light sources **520**B are arranged in the region on the −Y side and the +X side among the four regions divided by the straight lines P**2** and P**3**. The light sources **520**C are arranged in the region on the +Y side and the +X side among the four regions divided by the straight lines P**2** and P**3**. The light sources **520**D are arranged in the region on the −Y side and −X side among the four regions divided by the straight lines P**2** and P**3**.
(172) For each light source **520**A, there are light sources **520**B, **520**C, and **520**D whose centers C**2** are equally distanced from the axis of rotation D**3**. Thus, the light emitted by each light source **520**A and exiting the first lens **131** irradiates the same circular track irradiated by the light emitted by the light sources **520**B, **520**C, and **520**D whose centers C**2** are equally distanced from the axis of rotation D**3** and exiting the first lens **131**. This can mix light of at least two of four chromaticity in each irradiating region. Allowing the control unit **150** to control the output ratio of the light sources **520**A, **520**B, **520**C, and **520**D can adjust the degree of color mixing. This allows for emission of light that has been tuned to a predetermined color.
(173) FIG. **20** is a partial cross-sectional view of a variation of the state in which the lens is fixed.
(174) As shown in FIG. **20**, the first lens **131** may be fixed to the substrate **110** via a support portion **232**. In such a case, the first lens **131** rotates with the substrate **110**. As such, the first lens **131** may rotate with the substrate **110**.
(175) FIG. **21** is a schematic diagram showing a variation of the method of controlling multiple light sources.
(176) FIG. **21** shows an example in which there is no overlap between any adjacent irradiating regions among the irradiating regions **121***a*, **122***a*, **123***a*, **124***a*, **125***a*, and **126***a*.
(177) The control unit **150** controls the outputs of the light sources **120** in a similar manner to in the first embodiment.

(178) As shown in FIG. **21**, except for the first light source **121**, the light emitted by each light source **122**, **123**, **124**, **125**, and **126** and exiting the first lens **131** can be divided into multiple sections **151** on the circular tracks **122***b*, **123***b*, **124***b*, **125***b*, and **126***b*, respectively. In other words, the irradiating regions **122***a*, **123***a*, **124***a*, **125***a*, and **126***a* in the irradiated plane P**1** are divided into multiple sections **151**. In FIG. **21**, each portion of the individual circular tracks **122***b*, **123***b*, **124***b*, **125***b*, and **126***b* divided by broken lines which radially extend corresponds to a section **151**. The control unit **150** may be adapted to control the output of each light source **122**, **123**, **124**, **125**, and **126** in the multiple sections **151**.
(179) For example, the first circular track **122***b* is divided into eight sections **151**, the second circular track **123***b* is divided into 16 sections **151**, the third circular track **124***b* is divided into 16 sections **151**, the fourth circular track **125***b* is divided into 24 sections **151**, and the fifth circular track **126***b* is divided into eight sections **151**. The sections **151** are set to divide camera's photographing region **152** shown as a rectangle in FIG. **21**. Accordingly, for the circular tracks **122***b*, **123***b*, and **124***b* which are substantially entirely located within the photographing region **152**, the sections **151** are set to have substantially the same length. In contrast, for the circular tracks **125***b* and **126***b* which are partly located within and partly on the outward of the photographing region **152**, because the sections **151** are set to divide the regions in the photographing region **152**, the lengths of the sections **151** are not uniform. As shown by the circular track **125***b* in FIG. **21**, the sections of each of the circular tracks **125***b* and **126***b* located within the photographing region **152** are preferably set to have uniform lengths. However, the numbers and lengths of the sections in each of the circular tracks are not limited to those described above.
(180) The control unit **150** determines set output values for each of the light sources **122**, **123**, **124**, **125**, and **126** per section **151**. As the substrate **110** rotates, the sections **151** irradiated by the light emitted by the light sources **122**, **123**, **124**, **125**, and **126** and exiting the first lens **131** are switched, and the control unit **150** switches the outputs of the light sources **122**, **123**, **124**, **125**, and **126** to the set values that correspond to the sections **151** after switching.
(181) In the manner described above, the irradiating region **121***a* and the 72 sections **151** can divide the photographing region **152** into 73 brightness-adjustable regions.
(182) The constituent elements of the embodiments and variations described above can be suitably combined to the extent that such a combination does not cause inconsistency.
(183) The present disclosure includes the aspects described below.
(184) [Aspect 1]
(185) A light emitting module comprising:
(186) a substrate; a plurality of light sources fixed on the substrate and generating individually controllable outputs; a light shielding member disposed between the light sources; a first lens disposed above the light sources and on which the light emitted from each of the light sources becomes incident; and a drive unit capable of rotating the substrate, wherein the light sources include a first light source and a second light source, an angle formed by an axis of rotation of the substrate and an optical axis of the first light emitted by the first light source and exiting the first lens differs from an angle formed by the axis of rotation and an optical axis of second light emitted by the second light source and exiting the first lens, and the second light can irradiate a first circular track having the axis of rotation as its center.
[Aspect 2]
The light emitting module according to aspect 1 further including a control unit capable of controlling an output of the second light source in coordination with the drive unit.
[Aspect 3]
The light emitting module according to aspect 1 or 2 wherein the light sources include a third light source, third light emitted by the third light source and exiting the first lens can irradiate a second circular track having the axis of rotation as its center, and the second circular track is different from the first circular track.

[Aspect 4]
The light emitting module according to aspect 1 or 2 wherein the light sources include a third light source, third light emitted by the third light source and exiting the first lens can irradiate the first circular track, and a chromaticity of the third light differs from a chromaticity of the second light.
[Aspect 5]
The light emitting module according to any of aspects 1 to 4 wherein the first lens rotates with the substrate.
[Aspect 6]
The light emitting module according to any of aspects 1 to 5 wherein the first lens is a solid of revolution formed around the axis of rotation as a central axis, and an optical axis of the first lens coincides with the axis of rotation.
[Aspect 7]
The light emitting module according to any of aspects 1 to 6 wherein the light emission face of the first lens is flat.
[Aspect 8]
The light emitting module according to any of aspects 1 to 7 wherein the light incident face of the first lens is a convex face.
[Aspect 9]
The light emitting module according to any of aspects 1 to 8 wherein the first lens is a Fresnel lens in which the light incident face has a Fresnel profile.
[Aspect 10]
The light emitting module according to any of aspects 1 to 6 wherein the light incident face and the light emission face of the first lens are both convex faces.
[Aspect 11]
The light emitting module according to any of aspects 1 to 10 further comprising second lenses positioned between the first lens and the light sources, and disposed individually on the light sources.
[Aspect 12]
The light emitting module according to any of aspects 1 to 10 further comprising a second lens positioned between the first lens and the light sources, and disposed over and across all of the light sources.
[Aspect 13]
The light emitting module according to any of aspects 1 to 12 wherein, in a top view, the light sources are arranged at intersections of a plurality of first straight lines and a plurality of second straight lines, the plurality of first straight lines extend in a first direction and are arranged in a second direction orthogonal to the first direction, and the plurality of second straight lines extend in the second direction and are arranged in the first direction.
[Aspect 14]
The light emitting module according to any of aspects 1 to 12 wherein, on the substrate, the light sources are arranged on a plurality of circumferences around the axis of rotation having radii different from one another.
[Aspect 15]
The light emitting module according to any of aspects 1 to 12 wherein, in a top view, the light sources are arranged at intersections of a plurality of first straight lines and a plurality of second straight lines, the first straight lines extend in the first direction and are arranged in the second direction orthogonal to the first direction, and the plurality of second straight lines extend in a third direction intersecting the first direction and the second direction and are arranged in a fourth direction orthogonal to the third direction.
[Aspect 16]
The light emitting module according to any of aspects 1 to 12 wherein the light sources are

arranged in a honeycomb shape.
[Aspect 17]
The light emitting module according to any of aspects 1 to 16 wherein, in a top view, a shape of each of the light sources is a hexagon.
[Aspect 18]
The light emitting module according to any of aspects 1 to 17 further comprising a light transmissive member disposed over and across all of the light sources.
[Aspect 19]
The light emitting module according to aspect 18 wherein the light transmissive member contains a light diffusing material.
INDUSTRIAL APPLICABILITY
(187) The present disclosure can be utilized, for example, as a light source for a camera flashlight, lighting fixture, automotive headlight, or the like.
REFERENCE NUMERALS
(188) **100**, **200**, **300**, **400**, **500**, **600**: light emitting module **110**: substrate **110***a*: upper face **110***b*: lower face **111**: resin layer **112**: wire **120**, **121**, **122**, **123**, **124**, **125**, **126**, **220**, **320**, **420**, **520**, **520**A, **520**B, **520**C, **520**D: light source **120***a*, **220***a*: light emitting element **120***b*, **220***b*, **320***b*, **420***b*, **529**A, **529**B, **529**C, **529**D: wavelength conversion member **120***c*, **220***c*: semiconductor stack structure **120***d*, **120***e*, **220***d*, **220***e*: electrode **121***a*, **122***a*, **123***a*, **124***a*, **125***a*, **126***a*: irradiating region **122***b*, **123***b*, **124***b*, **125***b*, **126***b*: circular track **131**, **331**, **431**: first lens **131***a*, **331***a*, **431***a*: light incident face **131***b*, **331***b*, **431***b*: light emission face **132**, **232**: support portion **140**: drive unit **141**: motor **142**: shaft **150**: control unit **151**: section **152**: photographing region **161**, **261**: light shielding member **162**: light transmissive member **170**: rotary connector **171**: ring unit **171***a*: tubular body **171***b*: ring **172**: brush unit **172***a*: brush **172***b*: holder **191**: casing **191***a*: opening **192**: smartphone constituent element **193**: sealing member **494**: cover member **580**, **680**: second lens C**1**: center of substrate C**2**: center of light source D**1**: axis D**21**, D**22**, D**23**, D**24**, D**25**, D**26**: optical axis D**3**: axis of rotation E**1**: circle F**1**: focal point G**1**, G**2**, G**3**, G**21**, G**22**, G**23**, G**24**, G**25**: circumference L**1**: first straight line L**2**, L**22**: second straight line P**1**: irradiated plane P**2**, P**3**: straight line θ**1**, θ**2**, θ**3**, θ**4**, θ**5**, θ**6**: angle

Claims

1. A light emitting module comprising: a substrate; a plurality of light sources fixed on the substrate and generating individually controllable outputs; a light shielding member disposed between and in contact with the light sources, the light shielding member comprising a resin material and a light diffusing material; a first lens disposed above the light sources and on which the light emitted from each of the light sources becomes incident; and a drive unit capable of rotating the substrate, wherein: the light sources include a first light source and a second light source, an angle formed by an axis of rotation of the substrate and an optical axis of first light emitted by the first light source and exiting the first lens differs from an angle formed by the axis of rotation and an optical axis of second light emitted by the second light source and exiting the first lens, and the second light can irradiate a first circular track having the axis of rotation as its center.
2. The light emitting module according to claim 1, further comprising a control unit capable of controlling an output of the second light source in coordination with the drive unit.
3. The light emitting module according to claim 1, wherein: the light sources include a third light source, third light emitted by the third light source and exiting the first lens can irradiate a second circular track having the axis of rotation as its center, and the second circular track is different from the first circular track.
4. The light emitting module according to claim 1, wherein: the light sources include a third light source, third light emitted by the third light source and exiting the first lens can irradiate the first

circular track, and a chromaticity of the third light differs from a chromaticity of the second light.
5. The light emitting module according to claim 1, wherein the first lens rotates with the substrate.
6. The light emitting module according to claim 1, wherein the first lens is a solid of revolution formed around the axis of rotation as a central axis, and an optical axis of the first lens coincides with the axis of rotation.
7. The light emitting module according to claim 1, wherein the light emission face of the first lens is flat.
8. The light emitting module according to claim 1, wherein the light incident face of the first lens is a convex face.
9. The light emitting module according to claim 1, wherein the first lens is a Fresnel lens in which the light incident face has a Fresnel profile.
10. The light emitting module according to claim 1, wherein the light incident face and the light emission face of the first lens are both convex faces.
11. The light emitting module according to claim 1, further comprising second lenses positioned between the first lens and the light sources, and disposed individually on the light sources.
12. The light emitting module according to claim 1, further comprising a second lens positioned between the first lens and the light sources, and disposed over and across all of the light sources.
13. The light emitting module according to claim 1, wherein, in a top view: the light sources are arranged at intersections of a plurality of first straight lines and a plurality of second straight lines, the plurality of first straight lines extend in a first direction and are arranged in a second direction orthogonal to the first direction, and the plurality of second straight lines extend in the second direction and are arranged in the first direction.
14. The light emitting module according to claim 1, wherein, on the substrate, the light sources are arranged on a plurality of circumferences around the axis of rotation having radii different from one another.
15. The light emitting module according to claim 1, wherein, in a top view: the light sources are arranged at intersections of a plurality of first straight lines and a plurality of second straight lines, the plurality of first straight lines extend in a first direction and are arranged in a second direction orthogonal to the first direction, and the plurality of second straight lines extend in a third direction intersecting the first direction and the second direction and are arranged in a fourth direction orthogonal to the third direction.
16. The light emitting module according to claim 1, wherein the light sources are arranged in a honeycomb shape.
17. The light emitting module according to claim 1, wherein, in a top view, a shape of each of the light sources is a hexagon.
18. The light emitting module according to claim 1, further comprising a light transmissive member disposed over and across all of the light sources.
19. The light emitting module according to claim 18, wherein the light transmissive member contains a light diffusing material.
20. The light emitting module according to claim 1, wherein: each of the light sources comprises a light emitting element, and a wavelength conversion member located above the light emitting element, and the light shielding member contacts a lateral surface of the light emitting element of each light source and a lateral surface of the wavelength conversion member of each light source.